Applied Microeconometrics Home Assignment

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1 Introduction

This report replicates the empirical strategy outlined by Lee (2008) to estimate the causal effect of incumbency on electoral outcomes in United States House elections. The analysis employs a sharp regression discontinuity design (RDD) that exploits the deterministic rule assigning incumbent status based on vote share margins. The following sections detail the assignment mechanism, visual evidence of the discontinuity, parametric estimation strategies, and sensitivity analyses regarding baseline covariates.

2 Task 1: Treatment Indicator and Assignment Mechanism

The treatment variable `victory` serves as a binary indicator for whether the Democratic party secured incumbent status. This variable is constructed such that $\texttt{victory}_i = 1$ if the Democratic vote share margin of victory ($\texttt{difdemshare}_i$) exceeds zero, and $\texttt{victory}_i = 0$ otherwise. The sample comprises 11,795 observations with 6,696 treated units (56.8%) and 5,099 control units (43.2%).

The assignment mechanism follows a sharp regression discontinuity design where treatment status is a deterministic function of the running variable V, defined as the Democratic vote share margin. The threshold v_0 is set at 0, representing a tied election or a 50 percent two-party vote share. Formally, treatment assignment obeys the rule $D_i = 1[V_i > 0]$.

Lee (2008) establishes that causal inference in this context relies on the assumption that the density of the running variable is continuous at the threshold for each individual. While candidates influence vote shares through effort and resources, the final vote count contains an inherent random component arising from factors such as weather, turnout fluctuations, or measurement error. Consequently, conditional on candidate characteristics, the probability of winning an election decided by a narrow margin approximates a random draw. This implies that treatment status is as good as randomized in a local neighborhood of the threshold. A critical testable implication of this local randomization is that pre-determined baseline covariates must be balanced across the threshold. Any discontinuity in these variables would suggest manipulation of the running variable and invalidate the design.

3 Task 2: Local Averages and Visual Evidence

To visualize the relationship between the running variable and the outcome, the sample was divided into bins of width 0.01 ranging from -0.25 to 0.25. Within each bin, the mean Democratic vote share in the subsequent election (`demsharenext`) was calculated. Figure 1 plots these local averages against the bin midpoints.

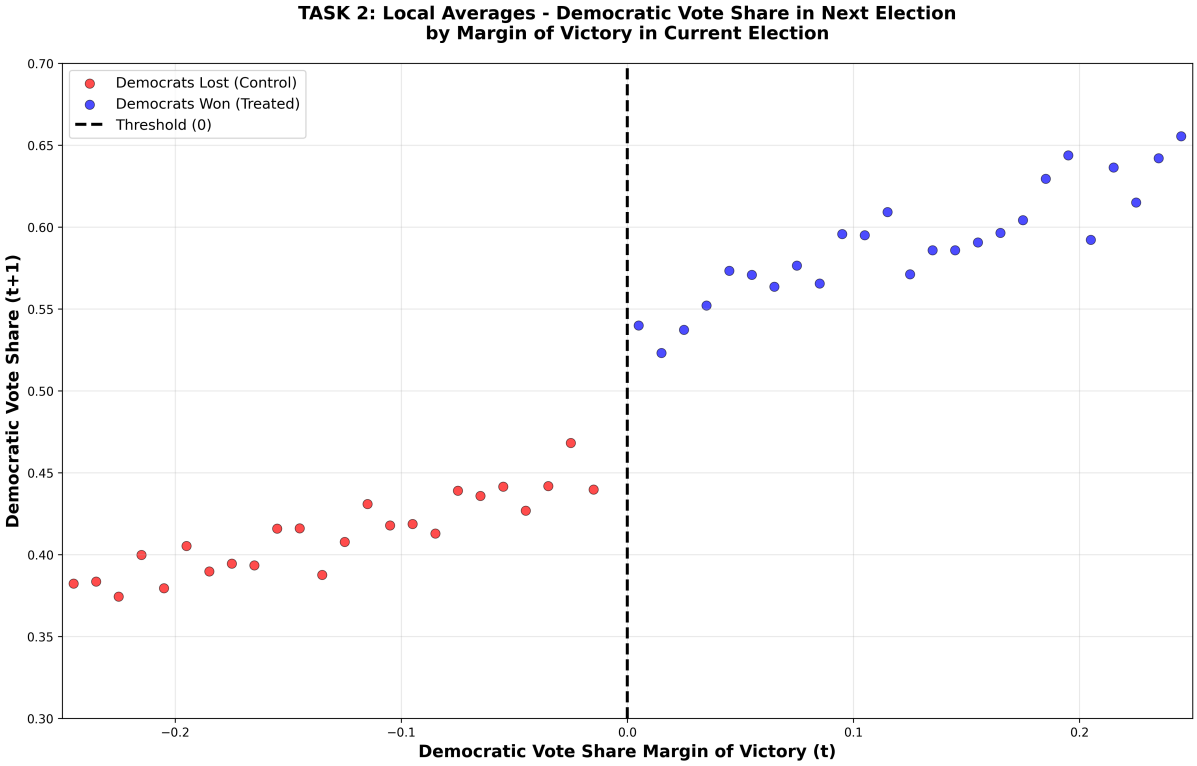

Figure 1: Local Averages of Democratic Vote Share in Election $t+1$ by Margin of Victory in Election t. The vertical dashed line indicates the threshold at 0.

The plot reveals a distinct discontinuity at the threshold. Districts where Democrats barely won exhibit substantially higher vote shares in the subsequent election compared to districts where Democrats barely lost. Specifically, the simple difference in means between the first bin to the right of the threshold and the last bin to the left yields a gap of approximately 10.03 percentage points. Beyond the jump at zero, the data display a positive slope on both sides, indicating that districts with stronger historical performance tend to maintain higher vote shares. This pattern suggests the presence of selection bias in raw comparisons but supports the validity of the RDD given the smoothness of the relationship within each group and the absence of other jumps.

4 Task 3: Polynomial Approximation and Discontinuity Estimation

To generate a precise estimate of the discontinuity gap, separate fourth-order polynomial regressions were fitted on either side of the threshold. The outcome variable `demsharenext` was regressed on a constant and powers of the running variable up to the

fourth order. This approach allows for flexible modeling of the underlying trend while isolating the jump at the cutoff.

Table 1 presents the estimated coefficients and standard errors for the polynomial terms. The intercepts from the left and right regressions represent the predicted vote shares at the threshold for losers and winners, respectively. The left limit is estimated at 0.4445 (standard error 0.0082), while the right limit is 0.5351 (standard error 0.0079). The difference between these limits constitutes the estimated incumbency advantage.

	Left Side ($V < 0$)		Right Side ($V \geq 0$)	
Coefficient	Estimate	Std. Error	Estimate	Std. Error
Constant	0.4445***	0.0082	0.5351***	0.0079
V^1	0.1561	0.1308	0.4817***	0.1202
V^2	-0.2175	0.6094	-0.3543	0.5210
V^3	1.2444	1.0244	0.5902	0.8123
V^4	1.4518***	0.5448	-0.3680	0.4074

Table 1: Fourth-Order Polynomial Regression Results. Dependent variable is Democratic vote share in election $t + 1$. Significance levels: *** $p < 0.01$.

The estimated discontinuity gap is 0.0906, or 9.06 percentage points, with a standard error of 0.0114. The associated t-statistic is 7.98, indicating statistical significance at the 1% level. Figure 2 overlays the fitted polynomial curves on the local averages, illustrating how the parametric model captures the non-linear trends while identifying the vertical shift at the threshold.

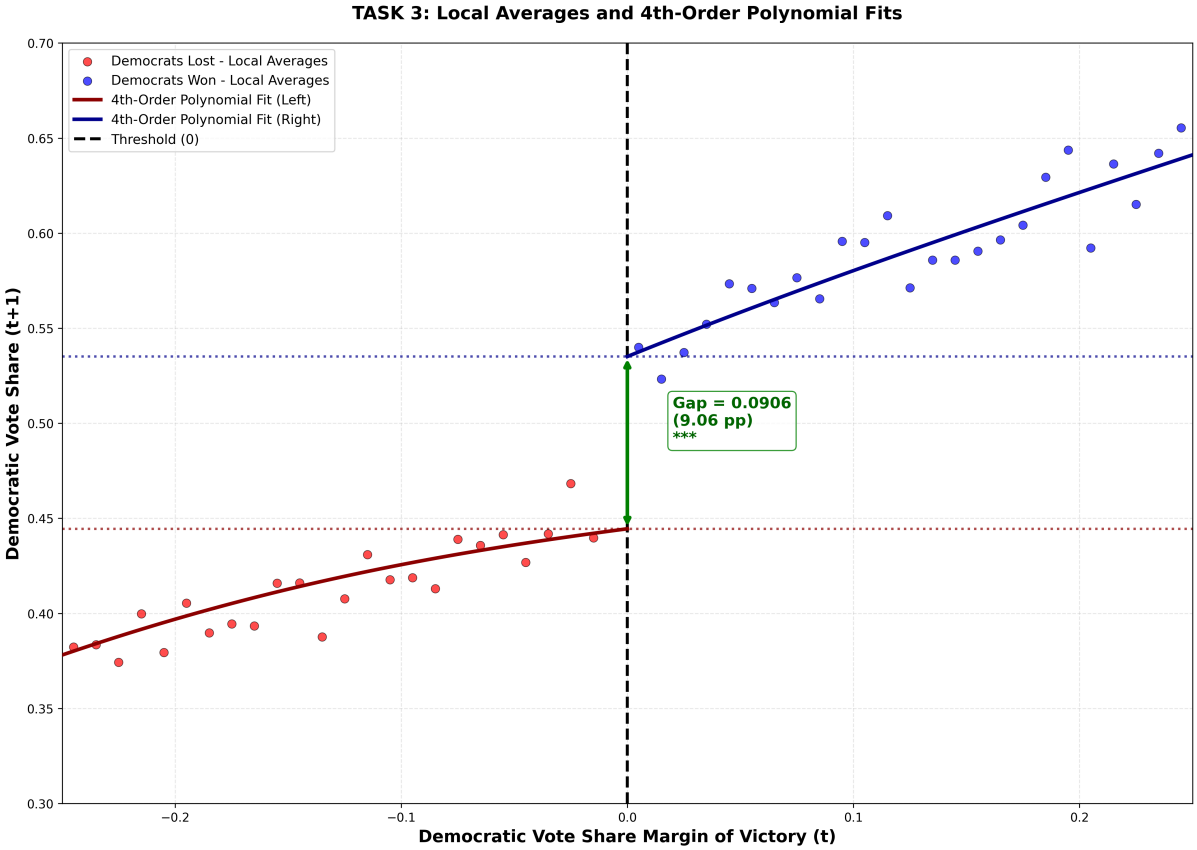

Figure 2: Local Averages and Fourth-Order Polynomial Fits. The green arrow indicates the estimated discontinuity gap of 0.0906.

5 Task 4: Unified Regression with Interactions

An alternative estimation strategy involves a single regression equation that includes the treatment indicator, the fourth-order polynomial in the running variable, and interaction terms between the treatment indicator and the polynomial terms. This specification allows the functional form to differ on either side of the threshold while directly estimating the treatment effect as the coefficient on the victory indicator.

The regression model takes the form:

$$Y_i = \alpha + \tau D_i + \sum_{k=1}^{4} \beta_k V_i^k + \sum_{k=1}^{4} \gamma_k (D_i \times V_i^k) + \epsilon_i \tag{1}$$

where Y_i is the vote share in election $t+1$, D_i is the victory indicator, and V_i is the vote share margin. The parameter of interest, τ, represents the discontinuity gap at $V = 0$.

Table 2 reports the full set of coefficient estimates. The coefficient on `victory` is 0.0898 with a standard error of 0.0114. This estimate implies that winning a close election increases the Democratic vote share in the subsequent election by approximately 8.98 percentage points. The result is highly statistically significant ($t = 7.87$, $p < 0.001$). The 95% confidence interval ranges from 0.0674 to 0.1121.

Variable	Coefficient	Std. Error	t-statistic
Constant	0.4454***	0.0084	53.16
Victory	0.0898***	0.0114	7.87
V^1	0.1688	0.1343	1.26
V^2	-0.1666	0.6265	-0.27
V^3	1.3197	1.0539	1.25
V^4	1.4881***	0.5607	2.65
Victory $\times$ V^1	0.3118*	0.1788	1.74
Victory $\times$ V^2	-0.1837	0.8085	-0.23
Victory $\times$ V^3	-0.7351	1.3209	-0.56
Victory $\times$ V^4	-1.8535***	0.6883	-2.69

Table 2: Unified RD Regression Estimates. Dependent variable is Democratic vote share in election $t+1$. Significance levels: * $p < 0.1$, *** $p < 0.01$.

Figure 3 displays the observed data points alongside the fitted values from this unified model. The graph confirms the visual presence of the discontinuity and demonstrates that the model fits the data well on both sides of the cutoff. The estimated gap of 8.98 percentage points is consistent with the separate polynomial estimates obtained in Task 3.

Figure 3: RD Regression Fit: Observed versus Fitted Values. The plot shows the discontinuity in fitted values at the threshold.

6 Task 5: Sensitivity Analysis with Baseline Covariates

Under the local randomization assumption, the inclusion of pre-determined baseline covariates should not alter the point estimate of the treatment effect, although it may improve precision by reducing residual variance. To test this implication, the baseline specification from Task 4 was augmented with various sets of covariates determined prior to election t. These covariates include the Democratic vote share and win probability in election $t-1$, as well as measures of political and electoral experience for both the Democratic candidate and the opposition.

Table 3 presents the incumbency effect estimates across five specifications. Column 1 reproduces the baseline result without covariates. Columns 2 through 5 sequentially add previous election outcomes, political experience variables, electoral experience variables, and finally all covariates simultaneously.

Specification	Effect	Std. Err.	t-stat	Sig.	95% CI	Obs
Baseline	0.0775	0.0132	5.86	***	[0.052, 0.103]	6,559
+ Prev. Outcomes	0.0784	0.0127	6.19	***	[0.054, 0.103]	6,559
+ Pol. Experience	0.0782	0.0126	6.19	***	[0.053, 0.103]	6,559
+ Elect. Experience	0.0782	0.0126	6.19	***	[0.053, 0.103]	6,559
+ All Covariates	0.0782	0.0126	6.19	***	[0.053, 0.103]	6,559

Table 3: Sensitivity Analysis: Incumbency Effect Across Specifications. Dependent variable is Democratic vote share in election $t+1$. Significance levels: *** $p < 0.01$.

The estimates remain remarkably stable across all specifications, ranging narrowly from 0.0775 to 0.0784. The standard deviation of the estimates is only 0.0003, representing a coefficient of variation of 0.42%. Figure 4 visualizes this stability, showing that the point estimates and their confidence intervals overlap substantially regardless of the covariates included.

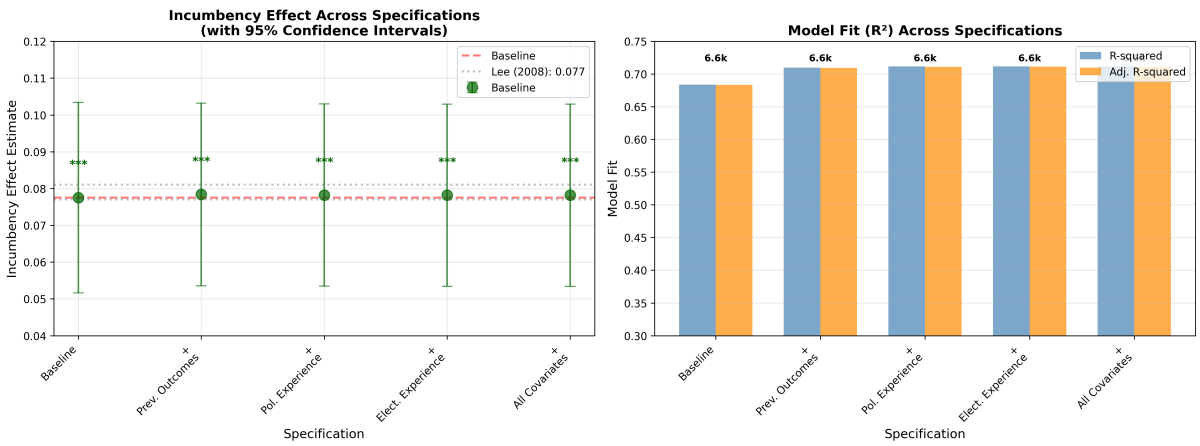

Figure 4: Incumbency Effect Estimates Across Specifications. The red dashed line indicates the baseline estimate, and the gray dotted lines indicate the range reported by Lee (2008).

This insensitivity to the inclusion of baseline controls provides strong supporting evidence for the validity of the regression discontinuity design. If the treatment assignment were confounded by omitted variables correlated with both the running variable and the outcome, one would expect the coefficient on the victory indicator to shift as relevant controls are added. The observed stability aligns with the theoretical prediction that treatment status is locally independent of pre-determined characteristics near the threshold.

7 Conclusion

The analysis confirms a substantial and statistically significant causal effect of incumbency on electoral performance in U.S. House elections. Across multiple estimation strategies, including local averages, separate polynomial fits, and unified interaction models, the estimated incumbency advantage consistently falls between 7.8 and 9.1 percentage points. The results are robust to the inclusion of extensive baseline covariates, reinforcing the credibility of the local randomization assumption. These findings replicate the core results of Lee (2008) and demonstrate that close elections provide a valid source of quasi-experimental variation for identifying causal effects in political economy.

References

References

[1] Lee, D. S. (2008). Randomized experiments from non-random selection in U.S. House elections. *Journal of Econometrics*, 142(2), 675–697.